

Problem Statement 2 – Output Guidelines

Package: SU007A – PCNL (Percutaneous Nephrolithotomy)

Purpose of the Outputs

Participants must generate **two outputs for each claim**:

S. No.	Output
1	Structured, row-wise output capturing: • Objective interpretation of pre-procedure and post-procedure radiological images (X-ray KUB / IVP) • Correlation with written radiology reports • Alignment with PCNL STG requirements • Stage and timeline awareness
2	Textual summary consolidating all structured outputs into a human-readable note suitable for PPD / CPD review, without making diagnostic, approval, or payment decisions

✦ **Note:** For PCNL, IVP is mandatory pre-procedure and X ray KUB is mandatory post-procedure.

SECTION 1: Claim Details *(1 column)*

Claim No.

- **Description:** Unique identifier for the claim
- **Type:** String / Integer
- **Purpose:** Links radiological interpretation to a specific PCNL claim

SECTION 2: Image Classification *(2 columns)*

1. Type of Radiological Image – Pre-Procedure Stage

- **Description:** Imaging submitted at pre-authorization stage
- **Expected Values:**
 - “IVP”
 - “Absent”
- **Purpose:** Confirms presence of mandatory stone-demonstrating investigation as per PCNL STG

2. Type of Radiological Image – Post-Procedure Stage

- **Description:** Imaging submitted at claim stage after PCNL
- **Expected Values:**
 - “X-ray KUB”
 - “Absent”
- **Purpose:** Confirms post-procedure evidence that PCNL was performed, as required under STG

SECTION 3: Image Description – Urinary Tract *(5 columns)*

Describe what is seen, not what it means

These fields capture **objective, observable image features** under the following five mandatory headings.

3A. Pre-Procedure Image Description

1. Pelvicalyceal System (PCS): Free-text, Appearance of the pelvicalyceal system on pre-procedure imaging; examples may include:

- PCS normal
- PCS mildly dilated
- PCS moderately dilated
- PCS bilaterally dilated
- PCS not well visualised

2. Ureter: Free-text, Appearance of ureters on imaging, examples may include:

- Normal
- Stone seen in left ureter / right ureter
- DJ stent visualised in ureter
- Ureter not visualised

3. Urinary Bladder: Free-text, Visual appearance of urinary bladder, examples may include:

- Normal
- Not distended
- Not well visualised

4. Stone Visualisation: free text, Presence and side of stone visualisation, examples may include:

- Visualised on left kidney
- Visualised on right kidney
- Visualised in ureter
- Multiple stones visualised (left / right / bilateral)
- Not clearly visualised

3B. Post-Procedure Image Description

5. Any Radio-Opaque Shadow or Calculus: Free-text, Presence of radio-opaque densities suggestive of calculi, examples may include:

- Yes (left kidney)
- Yes (right kidney)
- Yes (bilateral)
- No

6. DJ Stent / PCN Tube Visualised: Free-text, Evidence of procedure-related devices on post-procedure imaging, examples may include:

- DJ stent visualised on left side
- DJ stent visualised on right side
- PCN tube visualised
- No device visualised

SECTION 4: Report Validity and Correlation (for both pre and post procedure reports in this package) (8 columns)

4A. Pre-Procedure Report Validity

- 1. Intra-Report Consistency Check:** Whether the written report's observations and impression agree with each other; expected values:
 - Consistent

- Partially supported
- Unsupported
- 2. **Reason (Intra):** Free text, Examples:
 - “Observations and impression match”
 - “Stone mentioned in impression but not in findings”
- 3. **Image vs Written Report Correlation (Inter):** Whether image features support what the written report claims; expected values:
 - Consistent
 - Partially supported
 - Unsupported
- 4. **Reason (Inter):** Free text explaining support or mismatch; may refer to stone presence, location, or procedure evidence

4B. Post-Procedure Report Validity

- 5. **Intra-Report Consistency Check:** Whether the written report’s observations and impression agree with each other; expected values:
 - Consistent
 - Partially supported
 - Unsupported
- 6. **Reason (Intra):** Free text, Examples:
 - “Observations and impression advice”
 - “Written report not complete”
- 7. **Image vs Written Report Correlation (Inter):** Whether image features support what the written report claims; expected values:
 - Consistent
 - Partially supported
 - Unsupported
- 8. **Reason (Inter):** Free text explaining support or mismatch; may refer to DJ stent/ PCNL tube presence, location, or procedure evidence

SECTION 5: Claim, Package & STG Alignment (overall) (4 columns)

5A. Claim and Package Alignment

- 1. **Alignment Category:** Expected values -
 - Both pre-procedure and post-procedure investigations align
 - Only pre-procedure investigations align
 - Only post-procedure investigations align
 - None of the investigations align

2. **Reason for Alignment / Non-Alignment:** Free text, examples may include
 - “Post-procedure X-ray submitted to show PCNL done”

5B. Investigation and STG Alignment

3. **STG Alignment Outcome:** Expected values -
 - Consistent
 - Partially supported
 - Unsupported
4. **Reason for Deviation (mapped to PPD / CPD logic):** Free text, examples may include
 - “Post-procedure imaging does not show procedure evidence”

SECTION 6: Stage & Timeline Awareness (6 columns)

S. No.	Column	Description
1	Admission Date	DD/MM/YYYY
2	Pre-Procedure Investigation Date	DD/MM/YYYY
3	Post-Procedure Investigation Date	DD/MM/YYYY
4	Discharge Date	DD/MM/YYYY
5	Flag for one-month older documents for pre-procedure investigations	Yes / No
6	Flag before admission date or after discharge date documents for post-procedure investigations	Yes / No

SECTION 7: Quality (1 column)

- **Description:** free text, image usability comment, examples may include:
 - Good enough for human eyes
 - Poor quality
 - Overexposed / underexposed
 - Artefacts present

Summary PDF

- **Description:** Short human-readable case summary
- **Expected Content:**
 - Key image description summary
 - Report validity and correlation outcome
 - Package and STG alignment statement
 - Stage and timelines awareness summary of dates and flags

- **Tone:** Assistive, neutral, non-diagnostic

✓ Important Rules for Participants

- Use **observational language only**
- Do **not** confirm diagnoses (e.g., nephrolithiasis severity)
- Do **not** recommend approval or rejection
- Explicitly mention uncertainty if image quality is poor
- All fields must be populated (use “Not assessed / Not seen” where needed)